National University

of Computer & Emerging Sciences-Faisalabad

Syllabus: ↳ loops ↳ while loops ↳ logic ↳ if ↳ % (mode) logic

CS1002-Programming Fundamentals

Quiz_04 maheen

7.5

Total Time: 15 min **Total Marks: 10**

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Q1. **03**

Write down the output of the following code

```
#include <iostream>
using namespace std;

int main() {
        int a = 63, b = 13;
        int remainder;
        while (b != 0) {
                remainder = a % b;
                a = b;
                b = remainder;
                cout << b << ' ' ;
        }
        cout << "\nResult: " << a << endl;
        return 0;
}
```

163

b/3

63/

11

13

Output:
a=63 b=13
remainder = 11
a = 13
b = 11
remainder = 2
a = 11
b = 2
r = 1
a = 2
b = 1

r = 0
a = 1
b = 0

11 _ 2 _ 1 _ 0 _

~~Output~~

~~11 2 1 0~~

Output

11 2 1 0

Result: 1

3

11 2 1 0

Q2. **07**

Write a program that prints the counts of even and odd Harshad numbers separately between **[10-999]**.

Definition: A number is **Harshad** number if it is divisible by the **sum of its digits**. For example,

18 -> 1+8 = 9 -> 18 is divisible by 9 = 2, So 18 is Harshad number

21 -> 2+1 = 3 -> 21 is divisible by 3 = 7, So 21 is Harshad number

23 -> 2+3 = 5 -> 23 is not divisible by 5, So 23 is not Harshad number

```
#include< iostream>
using namespace std;
int main( ) {
  int n=10, digit, sum, countevn, countodd, temp
  ~~cout << "Sum~~
  while(n<=999) { temp = n
  while(temp>0)
  { digit = n% 10;
  sum += digit; }
  if (n% sum == 0){
        count even += 1; ?
  else.
        countodd += 1;
  n++;
```

Rough

evencount?

Actual Code →

Actual Code:

```
int main() {
int n=10, digit, sum, counteven, countodd, temp;

while (n<=999) {
  temp=n;
while (temp>0) {
  digit = temp % 10;
  temp /= 10;
  sum += digit;

  }
  if (n % sum ==0) {
      n % 2 ==0
      Count even += 1;
  }

  else n % 2 != 0
      countodd += 1;
  }
}
cout<< " counteven" << counteven <<endl;
cout << " countodd " << countodd <<end;
return 0;
}
```